# Network Security Milestone 2 Report

Constructing a Secure Multi-Branch Network Infrastructure

May 16,2026

# Contents


**1 Introduction** **3**

**2 Network Topology Overview** **3**
- 2.1 Branch 1 Network Design . . . 3
- 2.2 Branch 2 Network Design . . . 3

**3 IP Addressing Scheme** **4**
- 3.1 Branch 1 Addressing . . . 4
- 3.2 Branch 2 Addressing . . . 4

**4 Device Credentials** **5**

**5 Configuration and Verification Steps** **5**
- 5.1 Basic Device Initialization . . . 5
  - 5.1.1 Configuration Commands (on Branch 2 Edge Router) . . . 5
  - 5.1.2 Verification Commands . . . 6
- 5.2 VLAN Configuration on Branch 2 . . . 6
  - 5.2.1 Configuration Commands (on Core Switch) . . . 6
  - 5.2.2 Configuration Commands (on SW2) . . . 7
  - 5.2.3 Configuration Commands (on Switch3) . . . 7
  - 5.2.4 Verification Commands . . . 8
- 5.3 Interface Configuration and IP Addressing . . . 8
  - 5.3.1 Configuration Commands (on Branch 2 Edge Router) . . . 8
  - 5.3.2 Verification Commands . . . 9
- 5.4 OSPF Routing Protocol Configuration . . . 9
  - 5.4.1 Configuration Commands (Branch 2 Edge Router) . . . 9
  - 5.4.2 Verification Commands . . . 10
- 5.5 Configuring the VLAN . . . 10
  - 5.5.1 Configuration Commands (on Branch 2 Edge Router) . . . 11
  - 5.5.2 Verification Commands . . . 13
- 5.6 TACACS+ AAA Authentication . . . 13
  - 5.6.1 Configuration Commands (on each devices) . . . 13
  - 5.6.2 Verification Commands . . . 14
- 5.7 Syslog Logging . . . 14
  - 5.7.1 Configuration Commands (on each network device) . . . 14
  - 5.7.2 Verification Commands . . . 14
- 5.8 Save Configurations and Password Hashing . . . 14
  - 5.8.1 Configuration Commands . . . 14
  - 5.8.2 Verification Commands . . . 14
- 5.9 Port Security Configuration . . . 14
  - 5.9.1 Port Security Policy Design . . . 15
  - 5.9.2 Configuration Commands (Access Switches) . . . 15
  - 5.9.3 Verification Commands . . . 15
  - 5.9.4 Security Rationale . . . 15
  - 5.9.5 Port Security Deployment Table . . . 15

# 1 Introduction

This report documents the design and implementation of a secure, scalable multi-branch network infrastructure using Cisco Packet Tracer. The network consists of two branches: Branch 1 contains internal departments (Server Room, Technical Support, Employees' Offices) and a Demilitarized Zone (DMZ) for public-facing servers, while Branch 2 implements VLAN segmentation for different organizational units (Sales, Management, Marketing, IT) with centralized servers. Both branches connect through an external ISP connection. All devices are configured with OSPF dynamic routing, AAA authentication via TACACS+, centralised logging with Syslog, and various security controls including ACLs, IPS, and brute-force prevention.

# 2 Network Topology Overview

The network architecture consists of two main branches interconnected through core routing infrastructure:

## 2.1 Branch 1 Network Design

Branch 1 represents the primary office location with traditional departmental segmentation and a DMZ for external services.

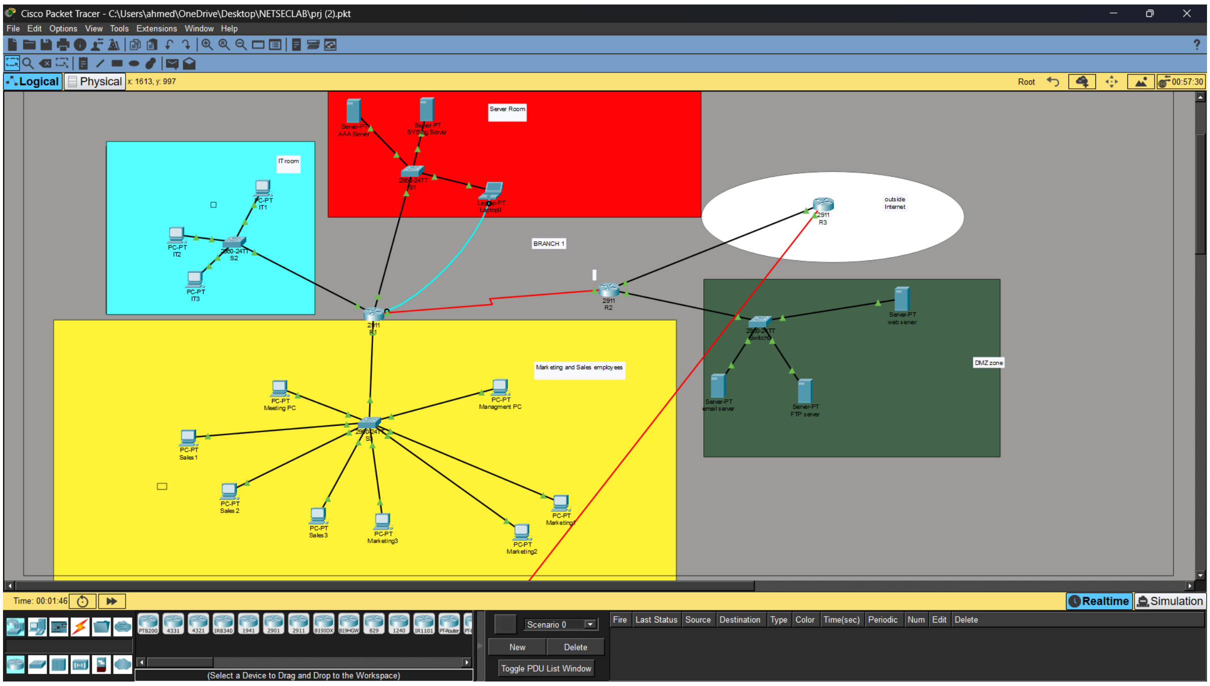



Figure 1: Branch 1 Network Topology

## 2.2 Branch 2 Network Design

Branch 2 implements VLAN-based segmentation for different organizational units, providing logical separation while maintaining centralized management.

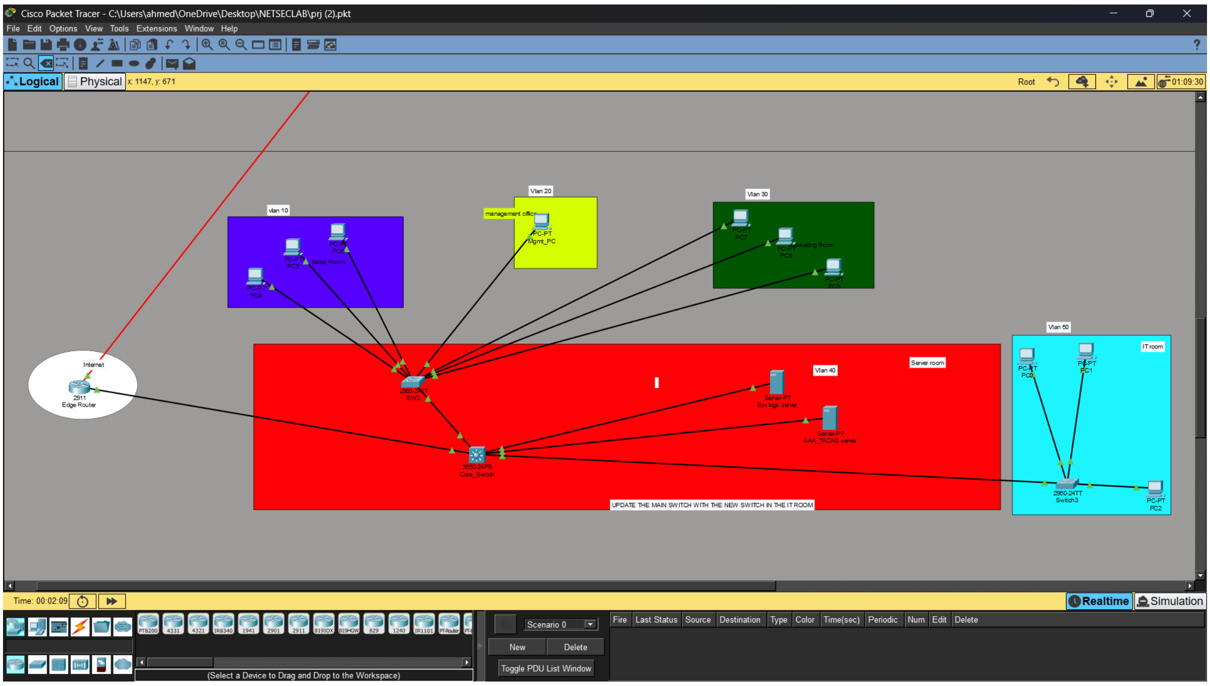


Figure 2: Branch 2 Network Topology

# 3 IP Addressing Scheme

## 3.1 Branch 1 Addressing

The network addressing scheme for Branch 1 is summarised in Table 1.

| **Device/Subnet** | **IP Address / Network** | **Notes** |
|---|---|---|
| Server Room | 192.168.1.0/24 | R1 interface: 192.168.1.1 |
| Technical Support | 192.168.2.0/24 | R1 interface: 192.168.2.1 |
| Employees' Offices | 192.168.3.0/24 | R1 interface: 192.168.3.1 |
| DMZ | 172.16.1.0/24 | R2 interface: 172.16.1.1 |
| R1–R2 Link | 172.16.5.0/24 | R1: 172.16.3.1, R2: 172.16.3.2 |
| R2–R3 Link | 172.16.2.0/24 | R2: 172.16.2.1, R3: 172.16.2.2 |
| Internet | 209.165.200.0/24 | R3: 209.165.200.1 |
| AAA Server | 192.168.1.10 | TACACS+ |
| Syslog Server | 192.168.1.20 | UDP 514 |
| Web Server | 172.16.1.30 | HTTP |
| FTP Server | 172.16.1.20 | FTP |
| Email Server | 172.16.1.10 | SMTP/POP3 |

Table 1: Branch 1 IP addressing scheme

## 3.2 Branch 2 Addressing

Branch 2 implements VLAN segmentation with the addressing scheme shown in Table 2.

| Device/VLAN | IP Address / Network | VLAN | Notes |
|---|---|---|---|
| Edge Router | 172.16.3.1 | – | Connection to core |
| Sales Room | 192.168.10.0/24 | VLAN 10 | |
| PC5 | 192.168.10.10 | VLAN 10 | Sales workstation |
| PC4 | 192.168.10.20 | VLAN 10 | Sales workstation |
| PC6 | 192.168.10.30 | VLAN 10 | Sales workstation |
| Management Room | 192.168.20.0/24 | VLAN 20 | |
| Mgmt_PC | 192.168.20.10 | VLAN 20 | Management workstation |
| Marketing Room | 192.168.30.0/24 | VLAN 30 | |
| PC7 | 192.168.30.10 | VLAN 30 | Marketing workstation |
| PC8 | 192.168.30.20 | VLAN 30 | Marketing workstation |
| PC9 | 192.168.30.30 | VLAN 30 | Marketing workstation |
| Servers | 192.168.40.0/24 | VLAN 40 | |
| Syslog Server | 192.168.40.10 | VLAN 40 | Centralized logging |
| AAA Server | 192.168.40.20 | VLAN 40 | TACACS+ authentication |
| IT Room | 192.168.50.0/24 | VLAN 50 | |
| PC0 | 192.168.50.10 | VLAN 50 | IT workstation |
| PC1 | 192.168.50.20 | VLAN 50 | IT workstation |
| PC2 | 192.168.50.30 | VLAN 50 | IT workstation |

Table 2: Branch 2 IP addressing scheme with VLAN assignments

# 4 Device Credentials

All network devices have been configured with secure passwords as detailed in Table 3.

| Device | Enable Password | Notes |
|---|---|---|
| Edge Router (Branch 2) | ER123 | Enable secret configured |
| SW2 (Branch 2) | SW2Pass | Enable secret configured |
| Core Switch (Branch 2) | CoreSwitchPass | Enable secret configured |
| Switch3 (Branch 2) | S123 | Enable secret configured |

Table 3: Device credentials for Branch 2 infrastructure

# 5 Configuration and Verification Steps

## 5.1 Basic Device Initialization

Before implementing advanced security features, all routers and switches must be configured with basic connectivity and secure management access. This section documents the initial setup procedures applied across both branches.

### 5.1.1 Configuration Commands (on Branch 2 Edge Router)

```
! Enter global configuration mode
configure terminal

! Set the device hostname
```

```
hostname EdgeRouter

! Configure enable secret
enable secret ER123

! Configure console line
line console 0
 login local
 exit
```

### 5.1.2 Verification Commands

```
! Check the running configuration for console settings
show running-config | section line con

! Verify device hostname and basic settings
show running-config | include hostname
```

## 5.2 VLAN Configuration on Branch 2

Branch 2 implements VLAN segmentation to logically separate different organizational units while maintaining centralized switching infrastructure.

### 5.2.1 Configuration Commands (on Core Switch)

```
! Enter configuration mode
configure terminal
hostname CoreSwitch

! Configure enable secret
enable secret CoreSwitchPass

! Create VLANs for different departments

! Configure VLAN interfaces for inter-VLAN routing
interface vlan 10
 ip address 192.168.10.2 255.255.255.0
 no shutdown
 exit

interface vlan 20
 ip address 192.168.20.2 255.255.255.0
 no shutdown
 exit

interface vlan 30
 ip address 192.168.30.2 255.255.255.0
 no shutdown
 exit

interface vlan 40
 ip address 192.168.40.2
 255.255.255.0
 no shutdown
 exit
```

```

interface vlan 50
 ip address 192.168.50.2 255.255.255.0
 no shutdown
 exit

! Configure trunk ports to access switches
interface gigabitEthernet1/0/1
 switchport mode trunk
 exit

interface gigabitEthernet1/0/4
 switchport mode trunk
 exit

 interface gigabitEthernet1/0/6
 switchport mode trunk
 exit
```

### 5.2.2 Configuration Commands (on SW2)

```
! Configure SW2
configure terminal
hostname SW2

! Configure trunk port to core switch
interface gigabitEthernet 0/1
 switchport mode trunk
 exit

! Configure access ports for workstations
interface range fastEthernet 0/2-4
 switchport mode access
 switchport access vlan 10
 exit

fastEthernet 0/5-7
 switchport mode access
 switchport access vlan 30
 exit

interface fastEthernet 0/1
 switchport mode access
 switchport access vlan 20
 exit
```

### 5.2.3 Configuration Commands (on Switch3)

```
! Configure Switch3
configure terminal
hostname Switch3

! Configure trunk port to core switch
interface gigabitEthernet 0/1
 switchport mode trunk
 exit
```

```

! Configure access ports
interface range fastEthernet 0/1-3
 switchport mode access
 switchport access vlan 50
 exit
```

### 5.2.4 Verification Commands

```
! Verify VLAN configuration
show vlan brief

! Verify trunk port configuration
show interfaces trunk

! Verify VLAN interface status
show ip interface brief | include Vlan
```

## 5.3 Interface Configuration and IP Addressing

### 5.3.1 Configuration Commands (on Branch 2 Edge Router)

```
! Configure interface to core switch
interface gigabitEthernet 0/1
 description Link to Core Switch
 ip address 192.168.5.1 255.255.255.0
 no shutdown
 exit

! Configure sub-interfaces for inter-VLAN routing
interface gigabitEthernet 0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.254 255.255.255.0
 exit

interface gigabitEthernet 0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.254 255.255.255.0
 exit

interface gigabitEthernet 0/1.30
 encapsulation dot1Q 30
 ip address 192.168.30.254 255.255.255.0
 exit

interface gigabitEthernet 0/1.40
 encapsulation dot1Q 40
 ip address 192.168.40.254 255.255.255.0
 exit

interface gigabitEthernet 0/1.50
 encapsulation dot1Q 50
 ip address 192.168.50.254 255.255.255.0
 exit
```

### 5.3.2 Verification Commands

```
show ip interface brief
show interfaces status
ping 192.168.10.10   ! verify connectivity to Sales PC
```

## 5.4 OSPF Routing Protocol Configuration

**Objective:** Configure OSPF on all routers in both branches to ensure full end-to-end connectivity.

### 5.4.1 Configuration Commands (Branch 2 Edge Router)

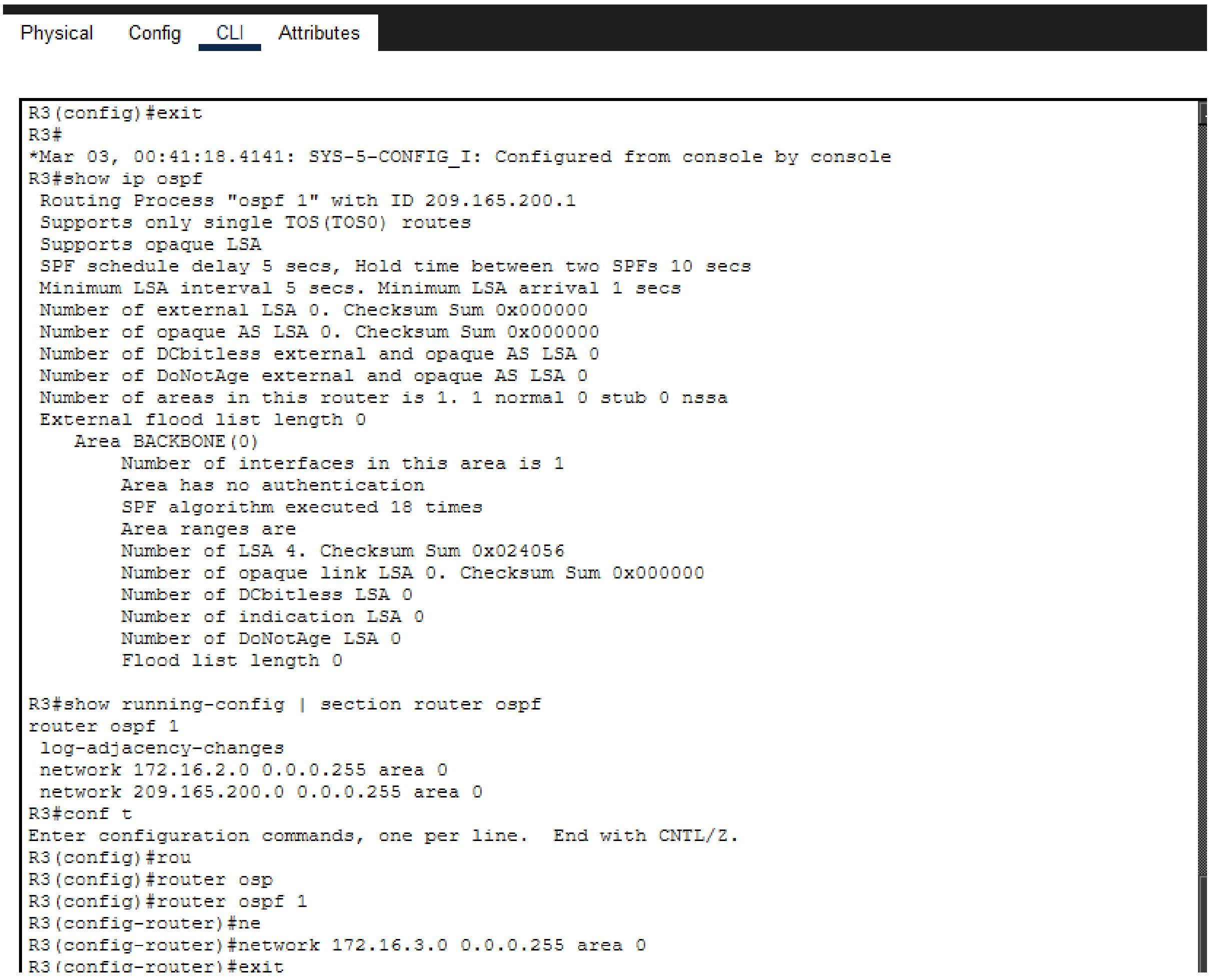
```
router ospf 1
 network 172.16.3.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 192.168.40.0 0.0.0.255 area 0
 network 192.168.50.0 0.0.0.255 area 0
```

Listing 1: OSPF configuration on Branch 2 Edge Router

```
router ospf 1
  network 172.16.3.0 0.0.0.255 area 0
```

Listing 2: OSPF configuration on R3

### 5.4.2 Verification Commands




```
R3(config)#exit
R3#
*Mar 03, 00:41:18.4141: SYS-5-CONFIG_I: Configured from console by console
R3#show ip ospf
 Routing Process "ospf 1" with ID 209.165.200.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        Area has no authentication
        SPF algorithm executed 18 times
        Area ranges are
        Number of LSA 4. Checksum Sum 0x024056
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

R3#show running-config | section router ospf
router ospf 1
 log-adjacency-changes
 network 172.16.2.0 0.0.0.255 area 0
 network 209.165.200.0 0.0.0.255 area 0
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#rou
R3(config)#router osp
R3(config)#router ospf 1
R3(config-router)#ne
R3(config-router)#network 172.16.3.0 0.0.0.255 area 0
R3(config-router)#exit
```

Figure 3: OSPF Neighbor Adjacency and Routing Verification on R3

```
show ip ospf neighbor
show ip route ospf
ping 192.168.40.10 source 192.168.10.10   ! from Sales PC to Syslog
    Server
```

Listing 3: Verify OSPF neighbors and routing table

## 5.5 Configuring the VLAN

**Objective:** Implement security policies between VLANs in Branch 2 to control traffic flow between departments.

### 5.5.1 Configuration Commands (on Branch 2 Edge Router)

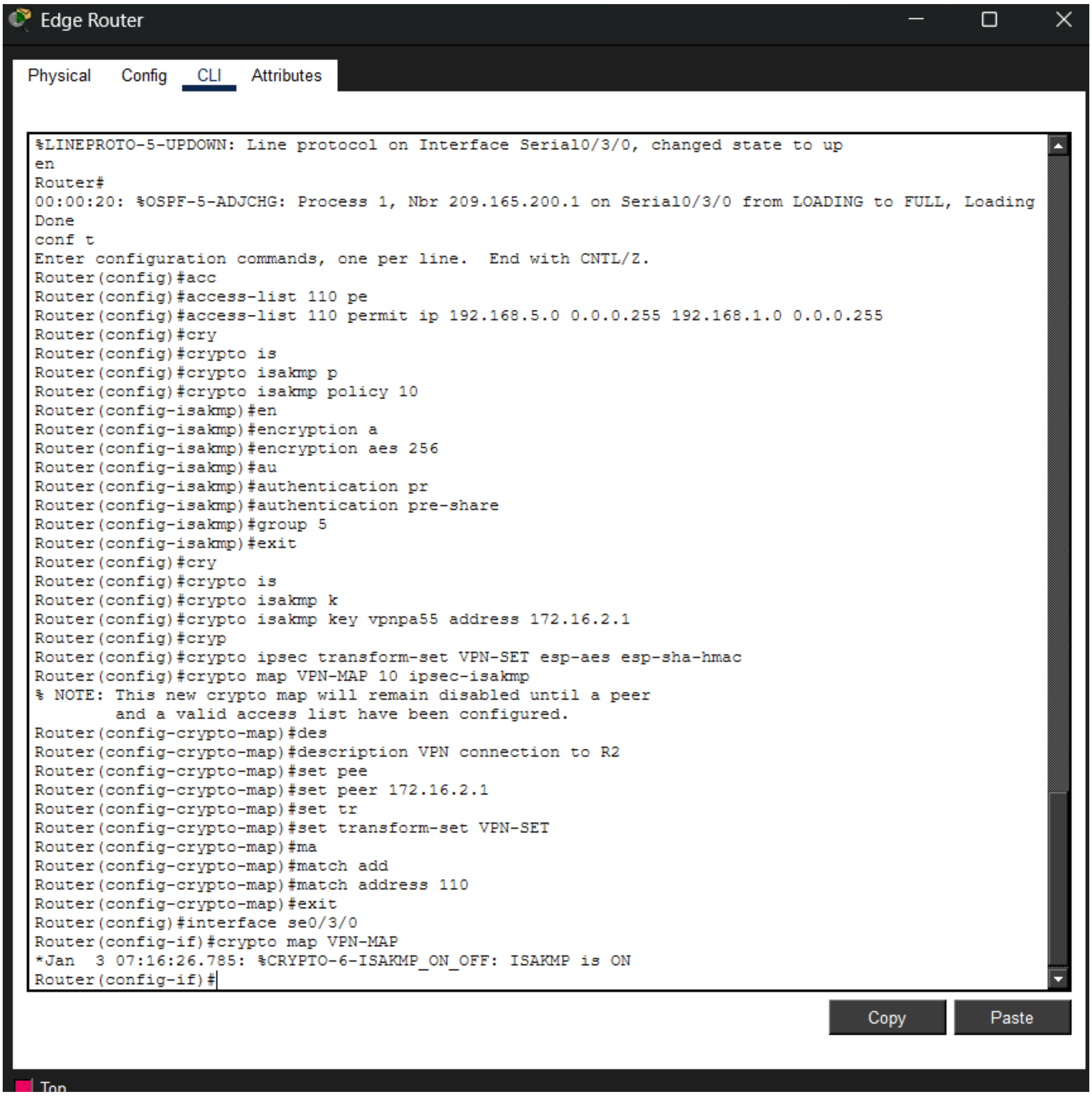



Figure 4: IPsec VPN Configuration on the Branch 2 Edge Router

```
! first install the license
license boot module c2900 technology-package securityk9

! Creating the ACL
! each preceded with access-list 110
permit ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
 permit ip 192.168.20.0 0.0.0.255 192.168.1.0 0.0.0.255
 permit ip 192.168.10.0 0.0.0.255 172.16.1.0 0.0.0.255
 permit ip 192.168.20.0 0.0.0.255 172.16.1.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 172.16.1.0 0.0.0.255
 permit ip 192.168.50.0 0.0.0.255 172.16.1.0 0.0.0.255

! Configure ISAKMP
crypto isakmp policy 10
```

```
encryption aes 256
authentication pre-share
group 5
exit
crypto isakmp key vpnpa55 address 172.16.3.2

! Configure IPsec
crypto map VPN-MAP 10 ipsec-isakmp
description VPN connection to R3
set peer 172.16.3.2
set transform-set VPN-SET
match address 110
exit

!Configure the crypto map on the outgoing interface.
interface se0/3/0
crypto map VPN-MAP

! Do the same in R3 but change 172.16.3.2 to 172.16.3.1
```

### 5.5.2 Verification Commands

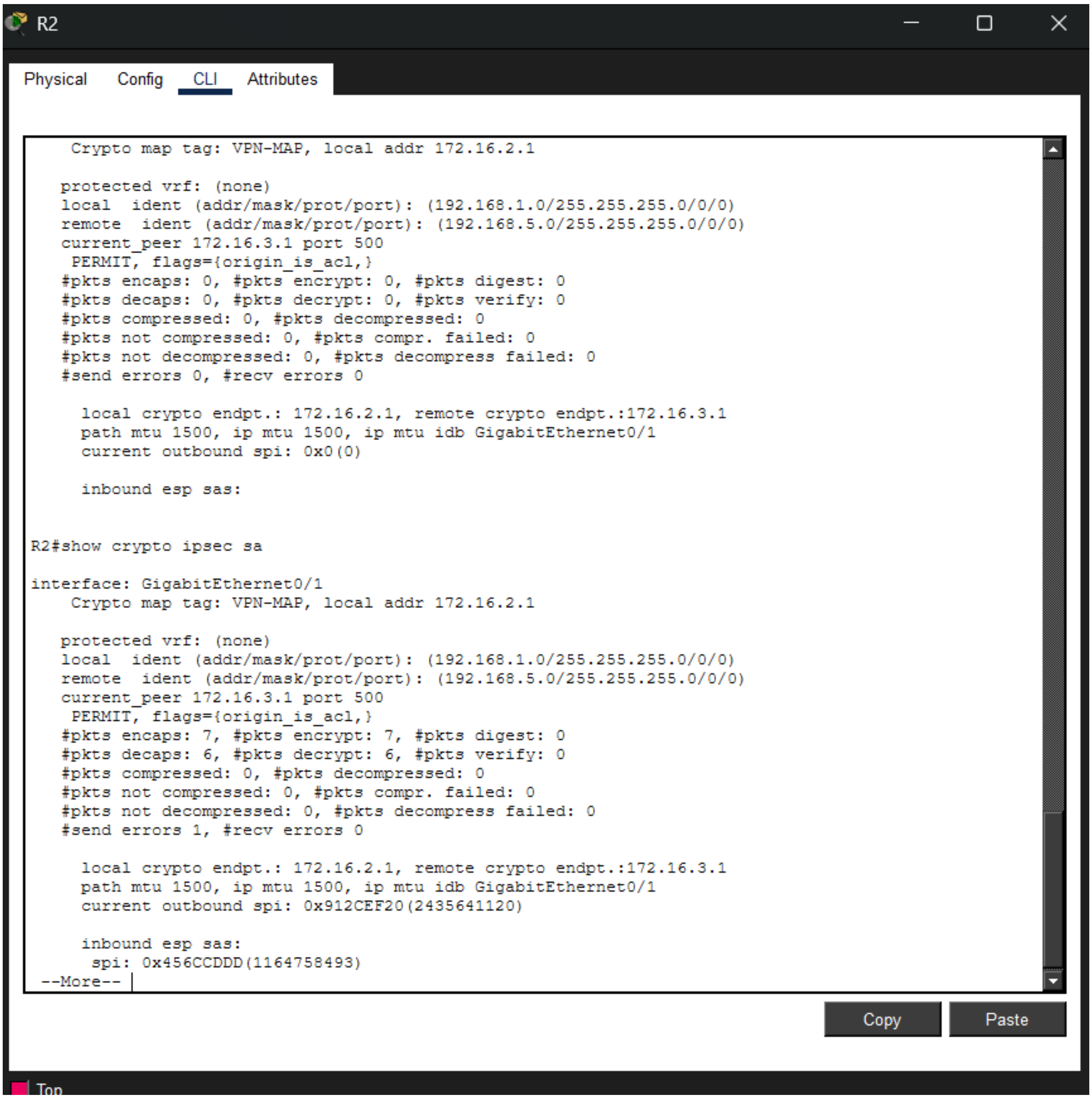



Figure 5: Verification of Active IPsec Security Associations Between R2 and R3

```
show access-lists 110
!use this command before and after a ping between edge router and R3
show crypto ipsec sa
```

## 5.6 TACACS+ AAA Authentication

**Objective:** Centralise authentication of all routers using a TACACS+ server.

### 5.6.1 Configuration Commands (on each devices)

```
aaa new-model
tacacs-server host 192.168.40.10 key cisco456
```

```
aaa authentication login default group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
```

### 5.6.2 Verification Commands

```
test aaa group tacacs+ SSHAdmin1 SSHadmin1Pa55 legacy
show aaa sessions
```

*Attempt SSH login to R1 using the remote credentials.*

## 5.7 Syslog Logging

**Objective:** Forward all system logs to a central Syslog server.

### 5.7.1 Configuration Commands (on each network device)

```
logging on
logging host 192.168.40.10
logging trap debugging
service timestamps log datetime msec
```

### 5.7.2 Verification Commands

```
show logging
! On Syslog server: observe incoming logs in the Services tab
```

## 5.8 Save Configurations and Password Hashing

**Objective:** Save running configuration to NVRAM and ensure all passwords are hashed.

### 5.8.1 Configuration Commands

```
enable secret class123
service password-encryption
copy running-config startup-config
```

### 5.8.2 Verification Commands

```
show startup-config | include enable secret
show startup-config | include password
```

## 5.9 Port Security Configuration

**Objective:** Implement Layer 2 port security on all access switches to prevent unauthorized devices, MAC flooding attacks, and ensure controlled endpoint access in the new branch network.

### 5.9.1 Port Security Policy Design

Port security was applied based on device criticality:

- **Shutdown Violation Mode:** Applied to critical infrastructure devices including AAA Server, SYSLOG Server, and Management PC. Any violation results in immediate port shutdown.
- **Restrict Violation Mode:** Applied to all user access ports (IT, Sales, Marketing PCs). Violations are logged and packets are dropped without shutting down the port.
- **Excluded Ports:** Trunk links between switches and the router are excluded from port security to maintain proper VLAN routing functionality.

### 5.9.2 Configuration Commands (Access Switches)

**Critical Devices (Shutdown Mode)**

```
interface fastEthernet 0/1
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation shutdown
```

**User Devices (Restrict Mode)**

```
interface range fastEthernet 0/2-7
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict
```

### 5.9.3 Verification Commands

```
show port-security
show port-security address
show port-security interface fastEthernet 0/1
```

### 5.9.4 Security Rationale

Port security ensures that only authorized devices can connect to switch access ports. Critical infrastructure ports are strictly protected using shutdown mode to immediately block unauthorized access attempts, while user ports use restrict mode to maintain network availability while logging violations for security monitoring.

### 5.9.5 Port Security Deployment Table

The following table summarizes the applied port security configuration across all access switches in the new branch:

| **Switch** | **Interface** | **Device** | **VLAN** | **Violation Mode** |
|---|---|---|---|---|
| Core Switch | gig1/0/3 | AAA Server | VLAN 40 | Shutdown |
| Core Switch | gig1/0/2 | SYSLOG Server | VLAN 40 | Shutdown |
| SW1 | Fa0/1 | Management PC | VLAN 20 | Shutdown |
| SW3 | Fa0/1 | IT PC1 | VLAN 50 | Restrict |
| SW3 | Fa0/2 | IT PC2 | VLAN 50 | Restrict |
| SW3 | Fa0/3 | IT PC3 | VLAN 50 | Restrict |
| SW1 | Fa0/2 | Sales PC1 | VLAN 10 | Restrict |
| SW1 | Fa0/3 | Sales PC2 | VLAN 10 | Restrict |
| SW1 | Fa0/4 | Sales PC3 | VLAN 10 | Restrict |
| SW1 | Fa0/5 | Marketing PC1 | VLAN 30 | Restrict |
| SW1 | Fa0/6 | Marketing PC2 | VLAN 30 | Restrict |
| SW1 | Fa0/7 | Marketing PC3 | VLAN 30 | Restrict |

Table 4: Port Security Deployment Across Access Switches